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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

5309229977

ENVIRONMENTAL MANAGEMENT **5014/23**

Alternative to Coursework **October/November 2010**

1 hour 30 minutes

Candidates answer on the Question Paper

Additional Materials: Ruler

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
Study the appropriate Source materials before you start to write your answers.
Credit will be given for appropriate selection and use of data in your answers and for relevant interpretation of these data. Suggestions for data sources are given in some questions.
You may use the source data to draw diagrams and graphs or to do calculations to illustrate your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use

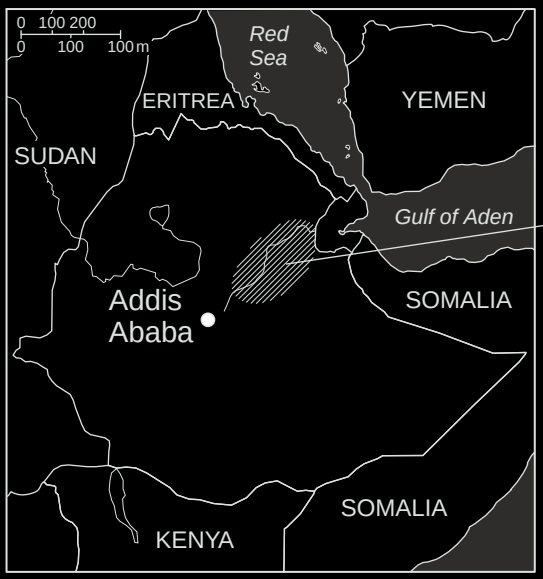
This document consists of **13** printed pages and **3** blank pages.

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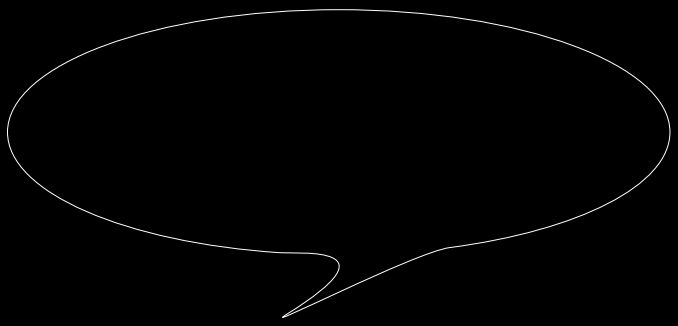


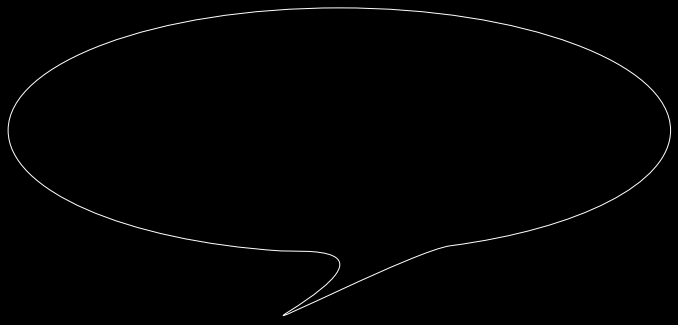
Map of Ethiopia



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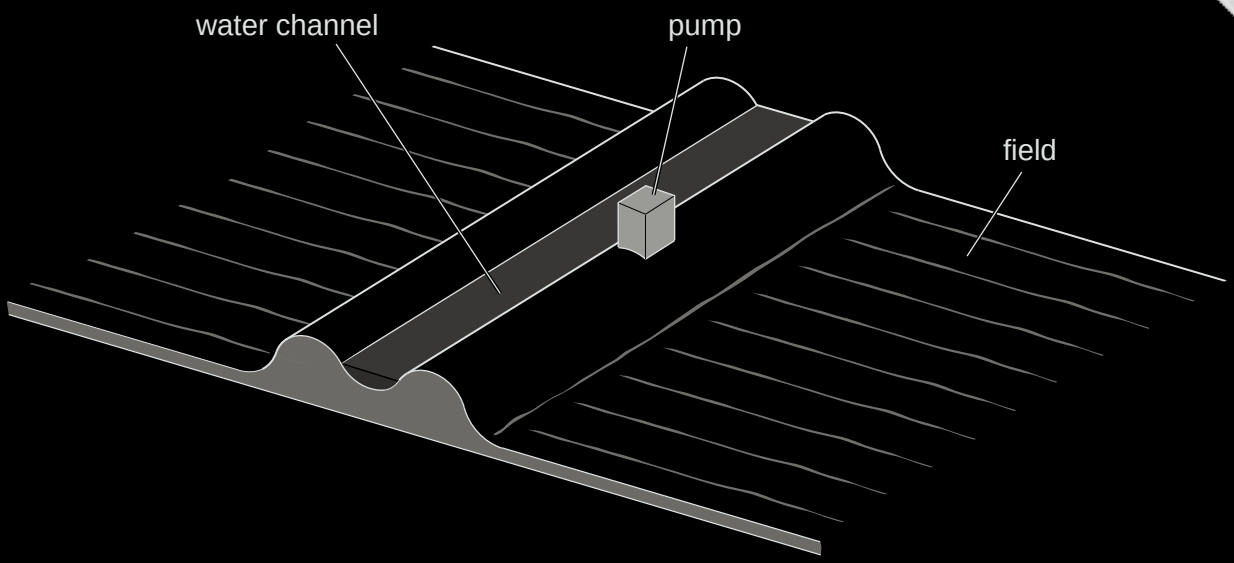


Fig. 1.1

(i) Complete **Fig. 1.1** to show a method for irrigating crops in the field. [2]

(ii) On irrigated land any yields decrease with time. Explain why.

.....

.....

.....

..... [3]

(iii) The farmer made this comment:

Since I have been irrigating my field the family has been suffering from malaria more often.

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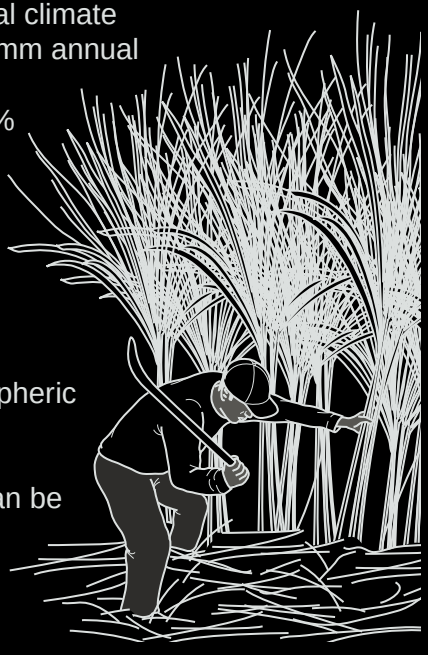
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Sugar Cane

- grows 2–6 metres in height
- grows in a tropical climate with at least 600 mm annual rainfall
- converts up to 2% solar energy into biomas.



Some sugar cane varieties fix atmospheric nitrogen.

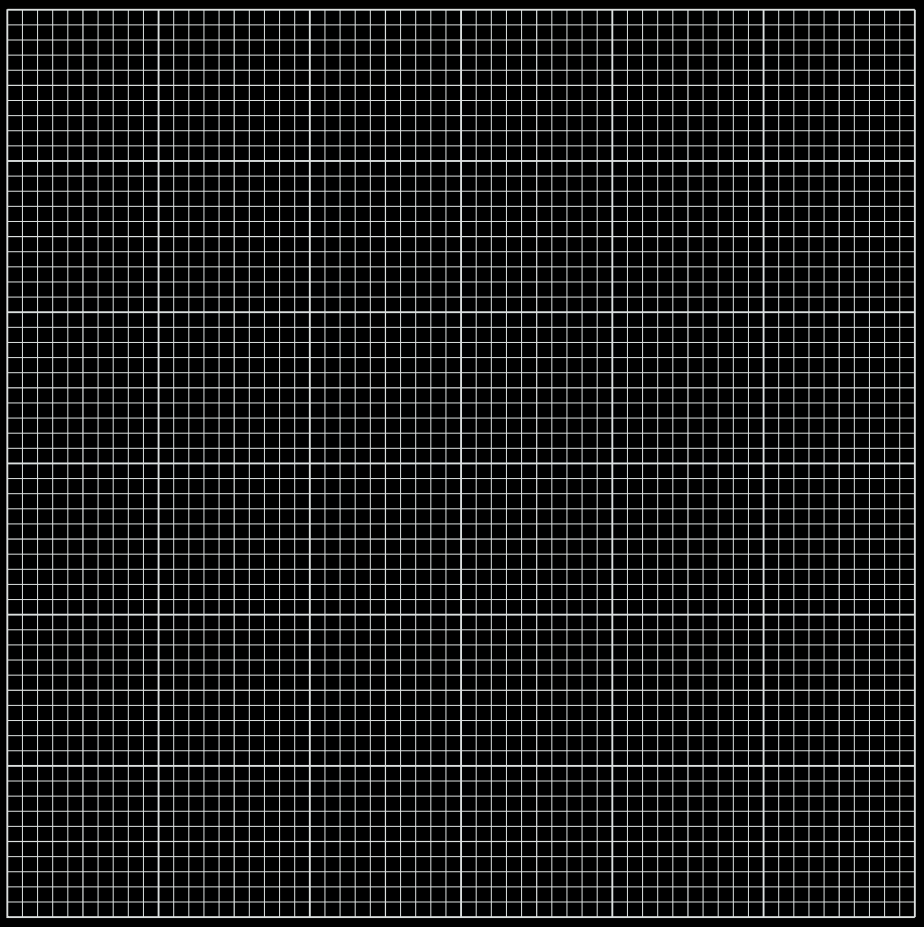
Waste cellulose can be made into paper.

The land is being cleared and planted. Many concrete irrigation channels are needed, as well as houses for the workers.

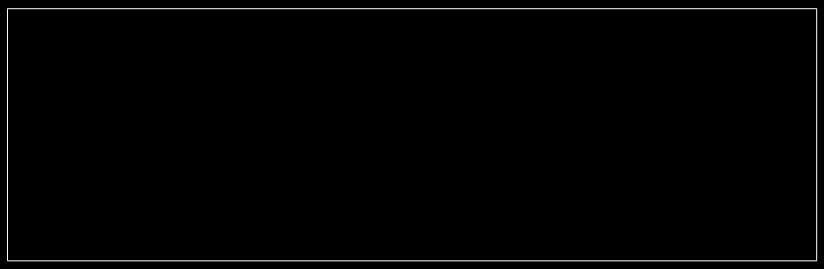
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